

Evaluating Plausibility in VR Beach Volleyball with varying levels of Crowd Population

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Virtual Reality)**

Supervisor: Dr. Gareth Young

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Declaration

I, the undersigned, declare that this work has not previously been submitted as an exercise for a degree at this, or any other University, and that unless otherwise stated, is my own work.

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...ABSTRACT...

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Chapter 1

Introduction

Athletes in professional, highly competitive sports have every interest in resorting to the newest advances in technology in order to enhance their competitive performance, and training regime. With recent jumps in technology, Virtual Reality (VR) is increasingly establishing itself as a valuable tool to professional athletes in training scenarios, thanks to its ability to display a wider variety of settings and activity, at a higher level of control than training grounds ever could grant on a regular basis.

Earliest adoptions of VR, or immersive technologies in professional sport include 3D recreations of matches for more immersive match reviews from either first-person, or 3-dimensional non-fixed viewpoints in the early 2010s (Farley et al. (2020)).

The 2022 inews article "Half of Premier League clubs use virtual reality to heal injuries and recreate matchday pressure in training" (Cunningham (2022)) highlights the increasing popularity of VR across the elite-level football, or soccer, scene, with half of the twenty English Premier League teams of the time applying VR scenarios for injury rehabilitation, recreation of matchday atmosphere and pressures, and other training and academy uses, which were left more vague.

The advantages of this technology are manifold: besides the proven ability to positively affect performance and motivation through various different factors interacting together, e.g. immersion (Ijsselstein et al. (2004)), competitiveness (Nunes et al. (2014)), digital cooperation (Irwin et al. (2012)), agency in selecting and adjusting the environment (Neumann et al. (2018)), or adaptive difficulty levels (Gray (2017), Richlan et al. (2023b)), it is a technology that allows for safer training conditions, exemplified with Ritter et al. (2023) showcasing that gymnasts practising on a beam at floor level - determined to be a safer training environment for the study than the physical facility featuring a beam raised above the floor - whilst wearing a Head-Mounted Display (HMD) experienced similar psychological conditions to the control group practising in the physical environment. VR is

today also applicable at a small scale, e.g. in local gyms, or in dedicated spaces at home or in training centres, and affordably through commercial-grade HMDs (Farley et al. (2020)).

With the increased interest, there is a rise in scrutiny of which ways the incorporation of VR can benefit athletes, with potential for use in sharpening motor skills such as movement and coordination, cognitive skills like perception on the pitch or positioning, mental skills to better resist match or score pressure, or tactical and strategic proficiency in helping to understand and apply directives from the coach (Richlan et al. (2023b)). Newer research, such as Ritter et al. (2023) and Gerwann et al. (2025), also promotes the introduction of Augmented Virtuality (AV), which introduces interaction with physical objects to provide or simulate haptic feedback whilst performing tasks in VR.

However, despite the widening adoption of the technology and increased research, the factors affecting performance in, and skill acquisition through VR are still a very young field of research that is yet to accurately study the aspects in a more granular and standardised fashion, as various reviews of existing studies into the topic of performance in VR sports applications, and their effect on competitive output, highlight (Neumann et al. (2018), Richlan et al. (2023b), Gerwann et al. (2025)).

Despite a notable extension of the range of sports and sport tasks studied between the reviews of Neumann et al. (2018) and Richlan et al. (2023b), the former of which advocated for exactly this extension beyond mostly aerobic workout activities, a sport that seems conspicuously absent from studies so far is Volleyball, alongside any of its variations, and probably for good reasons, as it requires dealing with many inherent limitations of VR application to sport, outlined in Richlan et al. (2023b).

That being said, there is ample evidence that the acquisition and refinement of complex skills in sport does not require a perfect recreation of the environment, and can be compartmentalised to fit specific purposes at a time, as evidenced by Farley et al. (2020)'s proposal of a system making use of a skateboard with VR environments, to practise surfing in dry conditions for example. Therefore, volleyball seems a suitable candidate for further investigation.

This dissertation therefore aims to stimulate further research into applying VR to volleyball applications by investigating whether, and in what way, simulated crowds meaningfully affect, both positively or negatively, users' sense of immersion or, more specifically, the illusion of plausibility that can make or break this feeling of immersion - itself a pri-

mordial factor in promoting the plethora of positive effects made possible by interactive VR.

Background knowledge required to grasp the various aspects of this study will be outlined before a selection and comparison of existing literature on, or related to, the subject in Chapter 2.

The subject of the study, and proposed solution theretowards will be detailed in Chapter 3, while the technical contributions made throughout this project are showcased in Chapter 4.

The proceedings of the study and its results will be looked into in Chapter 5, before closing this dissertation with a critical look at the study and its results, and suggestions for directions research building upon this paper might be worth taking in Chapter 6.

Chapter 2

State of the Art

In order to set this study apart from existing research into VR application to sport, it is fundamental to understand what angle this paper is approaching the question of plausibility and immersion from, as well as the impact it would aim to make if applied or researched further. Following this exposition, some notable studies that come close to the aims and methodology of this one will be introduced, then compared to each other and the aims of this study in the closing stages of this chapter.

2.1 Background

When dissecting the title of this paper, "Evaluating Plausibility in VR Beach Volleyball with varying levels of Crowd Population", there are certain terms that require highlighting and explaining, if one is to understand the aims and approach to this paper.

The first such term, in order of appearance, is "Plausibility", however it is a term that first requires understanding and clearly defining the next "contentious" term, VR.

2.1.1 Virtual Reality

The term Virtual Reality describes a system using modern hardware and computer rendering techniques to produce a realistic looking, digital world for multi-sensorial experiences (Farley et al. (2020)), controlled by a tracking system for input, and display system for output (Kulpa et al. (2015)). In the specific use of HMD-controlled VR, sometimes also referred to as immersive Virtual Reality (iVR) (Gerwonn et al. (2025)), the viewpoint within the virtual scene is updated based on the position and rotation of the display. Images displayed in such systems are stereoscopic (rendered for both eyes separately) to provide an illusion of depth (Vignais et al. (2015)).

For the remainder of this paper, and to interpret the title of this study from now on, VR will stand in for iVR, as a HMD-controlled system (a Meta Quest 3 device, to be specific) was used for this project.

2.1.2 Immersion, Illusion of Place, and of Plausibility

When defining the experience of being cognitively and emotionally invested in a non-physical scenario, be it relating to a well-written character in a novel, or making reflexive kicks whilst playing FIFA a little too seriously, most people would speak of being "immersed" in their material.

For the longest time, this sense was mostly a feeling of mental tunneling into these materials, that never really tried to actively blend out the physical environment surrounding an immersed reader, moviegoer, or other spectator and participant of various forms of media. With the increasing availability and practicality of iVR technologies, and notably affordable, commercial-grade HMDs, this sense of immersion was taken to a whole new level. VR hardware is able to produce high-quality virtual environments, that, when combined with a HMD, attempts to entirely phase out the visual environment, instead granting a complete vision of this digital space. Complemented with spatial audio technology, and perhaps in the near future, various devices providing haptic feedback, it is a system that aims to hijack the senses into a new reality.

This complete rewriting of immersion for this specific wing of technology has prompted debate around redefining its meaning to suit the newest flavour of the term.

In his paper on "Spirit of Place and Sense of Place in Virtual Realities" (Relph (2007)), geographer Edward Relph translates the terms Spirit of Place, defined as the distinct identity and identifiability of a physical location, and Sense of Place, described as "the ability to grasp and appreciate the distinctive quality of places", in other words the, notably positive, feelings elicited in a person toward a physical place, to more virtual contexts.

While the main point of that paper was to evaluate what a virtual sense and spirit of place sounds like - concluding that, just like real places, this sense and spirit of digital place will result from long-term interplay between world builders and their communities in creating distinct identities to strong places - the author plays around with the word 'presence' as a large factor in making virtual worlds feel more real, when cautioning about how the border between "real and [...] virtual worlds is already porous".

This sense of presence is in turn indirectly picked up in Mel Slater's "Place Illusion and Plausibility can lead to realistic behaviour in immersive virtual environments" (Slater

(2009)), in which the author presents a definition of the terms "immersion", "Place Illusion (PI)", and "Plausibility Illusion (Psi)" to describe the ways in which iVR can "transform place and even self-representation". These definitions were revisited in the same author's "A Separate Reality: An Update on Place Illusion and Plausibility in Virtual Reality" (Slater et al. (2022)), informed by a decade of further research into the subject.

Immersion is described as an inherent quantitative feature of hardware systems representing the capabilities of these systems at guaranteeing an immersive experience through so-called sensorimotor contingencies, referring to actions we carry out to collect information, e.g. turning our head to view the scenery from a different angle (Slater (2009)).

A certain system can be considered 'more immersive' than another if it has, for example, a higher framerate, larger Field of View (FoV), more ways to control the environment e.g. degrees of freedom on the headset, various tracking methods such as hand, eye, or full body tracking, or better tracking accuracy.

On the other hand, PI derives from the sense of presence tangentially mentioned by Relph, which is defined here as the "sense of being there in the environment depicted by the [VR] system". PI builds upon this sense of presence by implying this feeling is present despite the knowledge that it is not a physically anchored place (Slater (2009), Slater et al. (2022)): "It is the strong illusion of being in a place in spite of the sure knowledge that you are not there".

This more intangible definition makes it a subjective measure that is to be evaluated, according to the author, through more indirect assessments involving feedback from questionnaire and behavioural responses to the environment.

Finally, Psi is a novel addition to the subjective sense of immersion referred to as an introduction to this section, which essentially describes the user reacting to virtual events as if they were real and physical. The exact proposed definition is "the illusion that what is apparently happening is really happening[, even though you know for sure it is not[.]" (Slater (2009), Slater et al. (2022)), i.e. the illusion that digital events such as a ball flying towards the user are happening, and trigger authentic responses, despite the knowledge these events have no consequences on the physical world, in this case that there would be no feeling of contact and pain when the ball does hit the user's face.

Psi is a notion with a lot of nuance, as it involves the virtual environment responding to actions performed by the user, events within it being directed at the user, and meeting the expectations users would have of the same event happening in reality (Slater et al. (2022)), which makes it a very tricky sentiment to reliably capture and assess, and even

harder to provide universally and reliably due to differing, and at times even contradictory expectations around digital events.

As an example of how subjective this measure is, in a study performed to establish how to improve a VR recreation of a Dire Straits concert, participants were asked to reflect on the plausibility of the concert, and some highlighted disturbance at the mix of the crowd (being in one study too male-dominant, and in a second study too female-dominant), the animations of musicians being out of sync with the music being played, or even the absence of smokers amongst the crowd for historical accuracy of the concert being portrayed (Slater et al. (2023)).

In their revision of the definitions for PI and Psi, Slater et.al. state that both illusions being active results in VR users responding realistically to situations and events in VR. There are many more such factors that interplay with PI and Psi to enhance or negate subjective immersion, such as embodiment or the illusion of body ownership, describing the comfort and identifiability of users with their digital avatars (Slater et al. (2022)), or coherence, describing the reasonability, and predictability of environments (Skarbez et al. (2018)), which are however not directly relevant to this study, so will not be detailed any further.

2.1.3 Beach Volleyball

A quick word is needed to introduce volleyball, to give an overview of its main phases of play and certain types of ball interactions, to explain the technical implementation later.

The main distinction between classic, or indoor volleyball and beach volleyball, beyond the obvious change in setting and resulting changes in mobility and outfits, is the team size, which features two teams of six in classic volleyball, but downsizes to teams of two (FIVB (2026)) for its sandy counterpart, and a few minor variations in rules that are not worth going into for concerns of length, but along the gross lines applied to this study, the rules are more or less the same.

A play, or 'rally' is started by a player serving the ball - releasing it, then hitting it with one hand (either 'underarm': from below, upwards, or 'overarm': from above, forwards) to the opposite side of the net directly (FIVB (2026)).

Following a receive, both teams have to endeavour to land the ball on the ground in the opposite court half, or make the opposing team drop the ball anywhere other than the

own half. The official, international governing body of Volleyball identifies several valid ways to manipulate the ball while it is in play (FIVB (2026)):

- The **Dig** is a forearm pass performed by aligning a player's arms to form a platform, to bounce the ball off in a desired direction.
- The **Set** is an overhead pass performed with both hands aimed at completely changing the direction of the ball, usually to set up a spike to hit the ball over the other side. It is a particularly sensitive move as it requires very short contact with the ball to be not be ruled as grabbing and penalised, and requires much sensitivity to aim and weight properly.
- The **Spike** is a powerful, one-handed shot aimed at smashing the ball over the net.

Additionally, the ball is allowed to make contact with any part of a player's body, so long as it is neither grabbed, nor held for any length of time.

During a match, the team having won the last point gets the right to serve the next rally, and a point is awarded for every rally, based on where the ball makes contact with any object other than an active player or the net, and the last team to touch the ball before this contact. A team wins a point if either the opposing team last touches a ball that lands out of bounds, or if the ball lands on the ground in the opposing team's court, regardless of last touch (FIVB (2026)).

Beyond the very basic outline, there are many more rules at various levels of nitpicking such as a maximum of three touches per team before the ball has to be touched by the opponents, as well as more ways to interact with the ball that are not formally recognised by the FIVB as named moves, e.g. the 'poke' which involves lightly touching a ball near the net to lob a block, however as they were not included in the technical contributions and could be outlined over an entire book, and I am sure the page count was already daunting to begin with, any other rules are not worth further consideration for the purposes of this report.

Beach Volleyball was chosen for this study due to the initial plan to also incorporate opponents to play against participants, which would, with only three other players, reduce the load, both computational during runtime and for implementation, in employing them. Subjectively, the fact it also accommodates for nicer scenery may have helped sway the decision as well.

2.1.4 Crowd Population

The last concept to introduce is that of Crowd Population.

Crowds form a significant part of matchday pressure, representing the expectations and emotional involvement from outside the pitch being brought directly to athletes and the sporting staff in general. Performance under public scrutiny, be it public speech, artistic or sporting performance, is a widely studied subject under physical conditions, and as mentioned in Chapter 1, is already a factor motivating football teams to adopt VR technology in their training regimes Cunningham (2022).

Sports under COVID lockdown conditions have provided a unique case study to research the impact of crowds, or the lack thereof, on performances in sports that chose to pursue competitive matches behind closed doors, such as many professional football or soccer leagues in Europe, with studies (Leitner et al. (2023), Leitner and Richlan (2021), Richlan et al. (2023a)) aiming to explain various such discrepancies between pre-COVID and mid-COVID seasons. In particular, inspecting changes in home advantage - the perceived performance boost received from supporters at a team's home ground, the impact of the missing crowd on staff 'emotionality' during matches, and the impact of the missing crowd on match officials respectively.

The findings were that the various effects the crowds were perceived to provide before COVID, with no 'control scenario' to prove or disprove the perception - the boost in motivation on home soil, the stoking of emotional reactions in any direction - mostly dissipated, and change was particularly salient in referee behaviour, who described the games as having been less motivating and less focused than matches with full crowds, despite having been easier to referee, and with more compliant and agreeable players and staff (Richlan et al. (2023a)).

As such, the main factors that determine a crowd's impact can be boiled down to their size and density, and their level of involvement or emotional setting. Therefore, for the purposes of this study, Crowd Population will be defined as the amount and density of virtual spectators to a VR sports event, as well as their level of energy and reactivity to the happenings on the court or playing field.

2.2 ”Feasibility of training athletes for high-pressure situations using virtual reality”

In this study Stinson and Bowman (2014), the authors aimed to prove the viability of VR simulations for sport psychology training by subjecting football/soccer goalkeepers to a high-fidelity immersive simulation of a penalty shootout scenario, a particularly stressful match event. The specific goal was to measure whether the simulation could elicit a real sense of anxiety that could, in theory, then be practised away using this same environment.

The simulation environment bases on three independent variables: known anxiety triggers, field of regard - a measure adjacent to FoV describing the degree to which the display surrounds the user, and simulation fidelity.

The researchers concluded that VR sports applications could indeed elicit feelings of anxiety, both physiologically and in test results, although the findings are mitigated by showcasing an overall low level of anxiety despite the rise compared to the baseline scenario, and that the demographic did not present many people with a higher trait anxiety, meaning its generalisability is somewhat debatable, according to the authors. Additionally, all independent variables were determined to have contributed to the feeling of anxiety, although the influence of field of regard had an adverse effect than expected, with high FoR leading to decreased confidence in lower anxiety-inducing settings w.r.t. the other two parameters.

2.2.1 Plausibility Assessment

The central measure for this study was anxiety, through both quantitative or physiological and qualitative means. This does indirectly measure Psi, considering that if the simulation was experienced as inauthentic, it would not elicit anxious behaviours whatsoever, though the specific measurement 'fails' to keep track of other factors such as the accuracy of ball movements in a simulated space with bent optics, due to the use of a CAVE-like system.

2.2.2 Methodology

A major difference in fundamental approach is the VR system. This study makes use of a CAVE-like display system, which projects the virtual environment onto a cube-shaped room, with 3D glasses employed to preserve the illusion of depth.

Participants therefore have a very different sense of agency in this simulation, as they defended their goal with their full physical body. Behaviour correctness was established

through directional head tracking to establish whether a user reacted fast enough, and in the correct direction to theoretically save the penalty.

Anxiety was measured through physiological measures including heart rate, heart-rate variability, and galvanic skin response, as well as two standardised anxiety tests, STICSA and CSAI-2R.

25 participants completed the study, excluding pilot participants and those that did not finish the study, and were paid for their time alongside a bonus for good performance, thanks to a small price per save, totaling up to US\$30. Ages ranged from 22 to 32, and featured 9 female and 15 male participants, with various levels of experience at competitive levels, and in goalkeeping.

2.2.3 Studied Subject

The study aimed to demonstrate the viability of VR simulations for the psychological training of athletes, by showcasing its ability to elicit anxiety in users. This implicitly makes it a study of Presence as a whole, but particularly plausibility as laid out in the Plausibility Assessment section for this study.

The proposed project, in contrast, aims to assess plausibility more broadly to find out points of friction that could cause breaks in presence, as opposed to this proof-of-concept approach with a specific factor in mind.

2.3 "The sentiment of a virtual rock concert"

The paper "The sentiment of a virtual rock concert" (Slater et al. (2023)) builds upon a similar study two years prior, based around the same premise of establishing through participant feedback in what ways a digital recreation of a historic concert, here a 1983 live performance of "Sultans of Swing" by Dire Straits, can be improved upon to maximise enjoyment, or by proxy, the PI and Psi metrics.

While the first study was described as mainly focusing on that idea, this second study was informed by surprising reports and findings in the previous study, which prompted a few adjustments to see in what ways the feedback changed. Notably, swinging the surrounding crowd which was, in the first study, described as too male-dominant to being entirely composed of female avatars. Furthermore some improvements were made with regards to band animation accuracy, and the timing of various dancing and clapping an-

imations, all of which were also mentioned negatively during the first study.

During both studies, participants were sent a link to complete the study from home (taking place during COVID lockdowns), filling out a questionnaire for demographic data, then running the VR simulation on a HMD, which ran for around 10 minutes. They would be placed in the crowd, surrounded by digital avatars interacting with the music. In study 1 their behaviour was fully independent of the participant, in study 2, some participants randomly selected between Look and NoLook, the latter being a control group, would experience the avatars staring back at them for 1-3s if they were looked at. Following the simulation, participants were then asked to write an essay about their experience, taking into account the factors of the feeling of being at the concert, interactions with the crowd, specifically whether any helped immerse better, or took from the experience, then answering a few questions on a 1-7 Likert scale relating to copresence (social belonging within the crowd), the willingness to dance along with the crowd, and how close the experience felt to a real concert experience.

The findings largely confirmed those made during the first study, with participants now often being thrown off by being around too many women, both for the occasion of an 80s rock concert and in general, the staring being off-putting and even creepy at times, and still featuring some mild resistance at the band when looked at more closely. However, the reports in this second study seemed to highlight more simple discomfort than the reports of creepy and menacing male audience members. Participants also widely seemed to express a high willingness to dance along with the crowd, and the overall feedback reads a lot like "it's fine until you look at the details", resulting in the assessment that overall Psi is fairly high, and could be improved even more by featuring a more diverse crowd, that stares less, and with more eye for detail regarding things like timing for clapping and musical animation accuracy.

2.3.1 Plausibility Assessment

This study aims to directly estimate Psi levels in participants in order to confirm whether it elicits a reaction at all, that could be construed as a successful implementation, and the sentiment analysis employed as its main methodology for measuring is also used to verify whether this reaction is mostly positive or negative. Whilst the findings are used to improve the simulation itself, the conclusion of the second study is that, for this kind of applications that require a high level of positive Plausibility to be successful, developers

should involve users in the design to incrementally improve the experience, as this set of studies has done.

2.3.2 Methodology

In order to assess participants' experience, the authors have asked them to answer a form for demographic data before the experience, then write an essay of at least 150 words relating to their experience, before rating three factors describing specific parts of the experience on a Likert scale.

The goal of that study is similar in nature to the proposed study, namely to assess users' plausibility with the underlying purpose to improve on a simulated environment. The study also features a larger participant pool ($n=26$) than that both envisioned and actually assessed for this study ($n=14$), a participant count that is neither unsubstantial, nor so rich that the findings can be considered as widely generalisable as can be, although the participant pool is recruited through more professional channels than the researchers' entourages, seeing as participation was proposed to students in foreign programmes, and some influx was recorded from social media advertisements as well.

The sentiment analysis, particularly using and comparing several different implementations of it, seems a fitting method for the overall reaction, and yields many avenues for comparison and details as evidenced by this section being the largest by far in the report. It allowed the researchers to narrow down participants in clusters of experience, determining there to have been four distinct groups of experience, by the resulting essays.

2.3.3 Studied Subject

The largest difference to the proposed study is the subject of this related work, which was to improve subjective immersion in a virtual concert, meaning focus both in analysis and development was mainly posed on realism and comfort to make the experience as real as possible, when exploring avenues to improve on the simulation. This proposed study aims to explore Plausibility as a factor that is essential to applications in competitive sport, to avoid breaks in immersion that might take athletes out. Same studied metric but to a very different end.

There is some overlap to be found however in the fact the virtual audience was subject to much scrutiny throughout the study e.g. in highlighting the anxieties around gender-

dominance in both ways, though a dampening to this similarity has to be mentioned due to participants also being cast as spectators, rather than being the spectated ones, as they will be in this study.

2.4 ”Indifferent or Enthusiastic? Virtual Audiences Animation and Perception in Virtual Reality”

This study led by a research team around Yann Glémarec looks into configuring a virtual crowd of spectators attending a presentation or public speech made by a VR user, with the goal of providing authentic non-verbal behaviour in the virtual crowd for a given level of enthusiasm for the speaker or subject (Arousal), and attitude towards the subject or speaker (Valence).

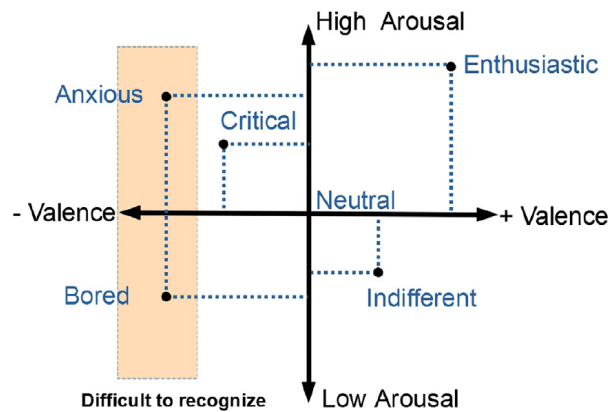


Figure 2.1: A graph showcasing the 2-dimensional scale used in the paper Glémarec et al. (2021) to showcase the relationship between arousal and valence in crowds. The points indicate named configurations used in phase 2 of the study, and the gray area represents an area participants had difficulty differentiating.

The study was conducted in two separate phases: in the first, twenty participants were asked to adjust a crowd’s collective posture, specifically gaze direction, the frequency of looking away from this direction, posture openness and proximity towards the speaker, types of head movements and facial expression, and their frequency. At the end of this adjustment game, participants were asked to rate their satisfaction with the result on a 1-7 satisfaction scale.

In the second phase of the study, 38 participants, none of which participated in the first phase of the study, were asked to hold a presentation in front of an audience generated with behaviour extrapolated from the results of the first phase of the study, and asked to

reconstruct the Arousal and Valence values associated with the exhibited behaviour. The crowds consisted of 10 avatars, 5 male, 5 female.

The motivation for this study was mostly therapeutic, in helping users become familiar with public speech, and overcome any anxieties or other stresses related to this situation.

The findings of the second study were that crowds with a higher valence, i.e. a more positive view of the speaker or presented topic, were better identified along the arousal scale, meaning that participants were well able to tell an enthusiastic crowd apart from an indifferent one. However there was more difficulty in differentiating arousal levels in crowds with negative valence, meaning that participants found it harder to differentiate between a bored and anxious crowd, each being descriptors at opposite ends of the neutral arousal values, as shown in figure 2.1.

In the guidelines made for audience simulation following the exposition of direct findings, the authors note that the main factors of interpretation for crowd sentiment was the gaze, both the direction and the frequency of change, head movements such as nodding or shaking the head, and the proximity towards the main actor, while noting that facial expressions should alternate between a neutral and a targeted expression to help impress the target, as a sustained facial expression might be interpreted as neutral otherwise, and would predictably be picked up on less if the spectator is further away.

An interesting note in the guidelines was also the authors reinforcing a finding from a different paper based on the results of phase 1, stating that a smaller concentration of negative actors could sway the overall sentiment to negative, stating that the sentiment could be considered positive if only 3 or 4 out of the 10 spectators acted negatively, whilst conversely it needed more than a majority of positive spectators for the sentiment to pass as positive.

2.4.1 Plausibility Assessment

Plausibility or immersion as a whole was not a direct measure of this study, though one could argue it may have been implicitly recorded with the rating of the crowds, since a user that does not believe the crowd to be realistic would not rate it appropriately.

2.4.2 Methodology

The results from the studies here were mostly directly taken from gameplay. In phase 1 of the study, the feature represented the vector of body movements embodied by the virtual crowd, as configured by the participant, paired with the associated valence and

arousal values it is to represent. In phase 2, the result is a pair of control V-A pair set by the system, or the 'real' V-A value, and the rating given by the participant, evaluated to compare the disparity.

This methodology showcases the efficacy of letting participants control the configurations to preference, as the evaluation of the configurations was left to be conducted by more participants, instead of imposing any interpretation of crowd sentiment by the research team, as stated by Murcia-Lopez et al. (2020), who conducted a larger study to assess various methodologies to assess plausibility, as a strong point towards this methodology.

2.4.3 Studied Subject

While this subject dives into assessing spectators as the main variable, the conducted study aims more to establish a reliable and recognisable crowd controller, that can accurately recreate the overall sentiment of a set of spectators. It also mainly regards their application in more professional or academic environments, however their findings may possibly be used for more types of crowds, including in sports, though the differences may warrant some dedicated study.

2.5 "Factors Affecting Sense of Presence in a Virtual Reality Social Environment: A Qualitative Study"

In this study aiming to inspect the factors affecting what the authors defined as Sense of Presence, a notion combining Slater's PI and Psi illusions under one hood, Riches et al. (2019) invited 76 participants to participate in a social VR experiment, where they joined a room of digital avatars having a party with the instruction to make an opinion of what the avatars thought of them. Following the scripted VR scenario lasting around 5 minutes, participants then took part in a semi-structured interview to extract the general feel of the experience.

Participants were recruited from a previous study based on scores in the paranoia trait on a Paranoid Thought Scales test, either scoring particularly high or low, and subject to other conditions such as not having a diagnosed history of mental health conditions, or other neurological disorders. 76 participants participated in total, 49 of which were

women, and with a mean age of 31.45 years.

The authors found that Presence was strongest when "the VR elicited genuine cognitive, emotional, and behavioral responses; and when participants created their own narrative about events [...]", i.e. when participants described a high sense of Psi. Inversely, Presence was lowest when participants "experienced diminished agency and physical impediments, such as cyber-sickness and awareness of apparatus and body movement".

In terms of the factors that affect Presence, the qualitative analysis of the interviews produced a list of themes that have had an impact on participants' presence, including but not limited to introspective thoughts and emotions, outwards-focused thoughts and emotions, physiological reactions, avatar behaviour, interactivity with the environment, and environmental characteristics, each with various subthemes that interacted differently with participants' presence.

2.5.1 Plausibility Assessment

In this study, the main subject of study was this notion of Presence, defined as a state of consciousness reliant on the perception of 'being there' in VR environments and as 'a psychological state in which virtual objects are experienced as actual objects in either sensory or nonsensory ways'. This definition mirrors quite accurately the combined definition of PI, the sense of "being there" [despite the sure knowledge it is not really the case], and Psi, the illusion that digital events are really happening.

By extension this means it indirectly does measure the cognitive effects of VR, but also incorporates PI in its assessment. As hinted in the description of the findings above, the observation was made that Psi had the greater positive effect on Presence, while the negative experiences were mostly attributed to degradation in PI, e.g. the awareness of the VR hardware, or sensory mismatches leading to cyber-sickness.

2.5.2 Methodology

Methodologically, this study joins the work of Slater et al. (2023) in running a purely qualitative assessment with verbal interviews that were analysed through Nvivo 11, a readymade tool to perform qualitative analyses of recorded interviews.

The strength of this study, compared to that of Slater et.al. is the sample size, which, with its count of 76, is almost three times larger than their 26, as described above, mean-

ing the findings can be described as more authoritative.

2.5.3 Studied Subject

The authors of this study approached the project from a therapeutic point of view, aiming to facilitate psychological assessment, and to demonstrate the viability of VR simulated scenarios for such purposes. Its findings can certainly be used for any immersive simulation, however the finding that the effect was strongest when it elicits "genuine cognitive, emotional, and behavioral responses" is akin to suggesting just 'doing it well'.

The outlined themes and subthemes identified from the interviews can be read like a guide for developers of what to look out for, and feelings or reactions to avoid or promote, which, while not a comprehensive guide of do's and don't's, can still be helpful in improving the starting point for any VR simulation.

2.6 Summary

In short, we have defined the study's title to be evaluating plausibility, the illusion that events in a VR environment are 'real' despite knowledge of the contrary, in an immersive virtual environment featuring a beach volleyball scenario, which will inspect the effect of changes in the density and involvement of a virtual crowd on the side.

In highlighting closely related works, we find that the studies into sports-specific adoption of VR, exemplified by Stinson and Bowman (2014), used Psi-adjacent concepts as proof-of-concepts for the viability of VR in the specific scenario, in this case the arousal of anxiety in participants.

Other studies more directly tasking themselves with the analysis of Presence or Psi, such as Riches et al. (2019) or Slater et al. (2023), have in the larger sense made use of qualitative methods to provide extensive insights into the underlying factors affecting this sentiment.

Finally, a study into the realism of crowd sentiment showcased a positive use-case for considering direct participant-controlled, level-based configurations directly in-simulation for assessment, whilst also highlighting useful insights into crowd behaviour and perception.

Having laid a solid foundation of information to understand the aims and proceedings of this project, the next chapter will build on the information outlined here to flesh out the identified research gap, and propose a research question and methodology to attempt to fill this gap.

Study	Plausibility Measured	Methodology Comparison	Subject
Stinson and Bowman (2014)	indirectly, and in limited fashion	physiological measurements + standardised tests	viability of VR for psychological training
Slater et al. (2023)	directly measured	sentiment analysis of essays + small questionnaire	improve on non-sporting simulation
Glémarec et al. (2021)	not measured	direct participant choices	crowd sentiment realism in behaviour and animation
Riches et al. (2019)	indirectly measured as part of Presence	qualitative analysis of semi-structured interviews	viability of VR for therapeutic purposes on mental conditions

Table 2.1: A comparison of the four presented closely-related works, with the ways Psi is measured, a quick overview of the employed methodologies, and the overall subject of study.

Chapter 3

Research Aims and Methodology

Having introduced the subject in a broad sense in chapter 1, and laid out background knowledge and existing works in chapter 2, it is now time to look forward and describe first the aims of this project, by way of identifying the research gap it aims to patch up, and the research question making up the substance of it. Following this, the proposed methodology will be laid out, alongside a description of the various components envisioned to make up a patch to fill the identified gap, that won't fly off under the first bit of scrutiny.

3.1 Research Gap and Research Question

The information laid out over the past two chapters, particularly the second, will be leveraged in order to outline all the edges and corners of the identified literature gap, then quickly outline a research question cut out to fit over it.

3.1.1 Identified Challenges

Chapter 2 established that existing use-cases of VR in high-level sport, as shown for Stinson and Bowman (2014), the question of plausibility was only tangentially addressed for the purpose of the project, which served more as a proof-of-concept than anything else. With Slater et al. (2023), and Riches et al. (2019), two approaches to measuring and improving the sense of subjective immersion, described as Psi in the former, and sense of presence in the latter, in order to make a simulated environment feel more lifelike, each resorting to detailed participant feedback in the form of essays or semi-structured interviews to extrapolate what should be done to improve the experience, each study leading to their own recommendations for best practises in the investigated scenarios.

Additionally, Glémarec et al. (2021), whose study served to flesh out a more expressive crowd in VR, illustrates, not only that certain features of body language pair in various

ways to exhibit widely different sentiments, but also that it neither takes a full body, nor even a full crowd, to impress these sentiments onto participants.

And finally, in the introduction (Chapter 1), the lack of meaningful such applications for use in volleyball was mentioned as well.

This absence is probably not down to chance, although there is no scientific evidence to back this up directly, considering that adapting volleyball to VR confronts developers with many limitations of the technology in development, as they are outlined in Gerwann et al. (2025)'s review, including but not limited to:

- the lack of haptic feedback to e.g. improve setting, which would be difficult to translate to real play at competitive levels,
- a reduced FoV which can lead to important peripheral information escaping perception,
- the limited ability to receive very precise physical cues in a timely manner, exemplified in the above review how batters in baseball need to rely on the pitcher's kinematics to predict ball speed, rotation and curve before it goes flying,
- the sheer complexity and granularity of some movements, in particular the set that would either need expensive programming to map accurately, paired with robust hand-tracking technology for use on professional athletes,
- the use of complex, and potentially hazardous movements in smaller spaces, e.g. diving for the ball in a controlled manner, or a very specific run up to jump when spiking in order to maximise jump height.

In this same review, the author emphasizes through these limitations that VR will never be, and never should be, a full substitution of training in physical environments, and should only be used where it makes sense to use for a real benefit. As a follow-up, the author explains that the potential for VR is highest towards injury prevention and the sharpening of mental skills, e.g. subjecting athletes to matchday pressures, or particularly stressful situations such as goalkeeping training under penalty shootout conditions in football/soccer Gerwann et al. (2025).

The aim of this research project being to contribute towards adoption of VR to enhance volleyball training regimes, it is the hope of this researcher to shorten development and design time for such training applications and scenarios with projects such as this one, by proposing best practises to form better starting points than guesswork on developers'

sides.

Considering that crowds form a large part of pressure during matches, it is proposed to look into crowds more specifically, in order to find out in what ways virtual crowds may interact with participants, to be able to suggest potential best-practise configurations to support Psi during gameplay, and, by extension, to keep motivation and performance up while immersed in VR training scenarios.

3.1.2 Proposed Work

As such, the question of what the study must investigate amounts to:

- assess participants' level of Psi during gameplay,
- determine the level of comfort and proficiency with the volleyball mechanics, to mention it as a potential factor in plausibility breaks.,
- provide different levels of crowd involvement to provide viable best practise estimations.

All of this subtly builds up to the research question this paper poses:

"Do trade-offs in environment population, such as variations in crowd population affect the illusion of plausibility in a VR Beach Volleyball setting?"

It is a question that may be tentatively answered by a few guiding hypotheses, such as

- The VR Beach Volleyball game is suitable as a baseline to establish a state of presence, including both PI and Psi.
- Users are fundamentally affected by the crowd in one way or another - i.e. the crowds are plausible.
- The digital crowds can reach a level of activity considered comfortable for users to play with.

Furthermore, a side hypothesis of "Users are more comfortable with a crowd than without one" can be proposed as well. In order to verify these hypotheses, a methodology is proposed in the following section.

3.2 Research Design

In order to look into the established research question and accompanying hypotheses, and considering it involves insights into participants' thinking processes, the experiment will consist of a task the recruited volunteers will be asked to perform, in order to gather data and an opinion to be evaluated towards establishing key findings.

In this section, we will first lay out a rough description of this above task, particularly how it will relate to participants' contribution towards research data, as the technical implementation will be outlined in much more detail in Chapter 4. Following the task description, a short overview of the hardware used for experiments will be presented before introducing the participants recruited for this study, as well as the procedure undergone for every individual trial.

3.2.1 Task

The task for the application was to play in a beach volleyball setting in VR. As soon as the first ball, always prepared for a serve, is played, participants are allowed to play a theoretically unlimited amount of points, regenerating after the previous ball is 'killed', i.e. hits an obstacle, or exits the bounds of the simulation space.

Every point starts with either a serve, or a ball played in from the other side of the court towards the player. There is a 1/3 chance to serve, and a 2/3 chance of a ball being played in, determined using a pseudo-random number generator for every point.

The methodology for the task is inspired by a recommendation outlined in Murcia-Lopez et al. (2020), to let participants influence the environment to their preference instead of either imposing or suggesting a qualitative ranking along the lines of 'configuration A is better than B' without a valid reason to do so. This is motivated by the finding that a more detailed configuration may lead to a decrease in plausibility due to e.g. failing expectations paired with the increased realism.

Therefore, the main variable given to participants to adjust was chosen to be crowd density. Four different levels were provided, increasing in count with rising density. Of a full, generated stand, each level $n \in [0, 3]$ showed every n^{th} participant out of 3, i.e. with a crowd filling of 0%, 33%, 67%, and 100% respectively. The crowds also produced a different audio track for every level, that increased in perceived crowd size with every level.

Participants enter the VR simulation on the highest setting, to get used to the environment, and start play in a 'free play' mode, in order to get used to VR and gameplay mechanics. During free play, the crowd always stayed at level 3, and points were not

counted.

Once they exit free play, they are encouraged to first explore all the levels, then adjust the crowd to their preference afterwards. It was possible for participants to stay at a level they liked for as long as they liked. Participants were also asked to end the experiment on the crowd configuration they preferred, all things considered. A prompt to adjust the crowd, in form of a UI field at eye level, appeared after every 5th point recorded, and only showed the change in crowd after the next level was selected and confirmed.

Points were counted and displayed in the environment to provide an incentive to perform well, although the fact performance was neither recorded nor rewarded was made clear to participants. This means that points served mainly as a stimulant for more competitive users, and a performance feedback mechanism for everyone else.

A history of crowd states is recorded for every participant, and will be used in the results discussion to outline findings.

In total, participants are asked to play at least 20 minutes including free play, in order to produce a large enough sample of transitions.

3.2.2 Hardware

All participants completed their trial using a standalone Meta Quest 3 device provided by the researcher, featuring a resolution of 2064x2208 pixels per eye, with a horizontal FoV of 103.8°, and a refresh rate of 120 Hz. The headset was complemented with a pair of Meta Quest Touch Plus controllers.

The study was conducted exclusively in the SCSS Computer Labs at Trinity College Dublin, in enclosed rooms rearranged to maximise the space available to participants, which gave approximately 3.5m by 4m ($\sim 14m^2$) of unobstructed space to participants. This was made to allow some flexibility of movement, notably for swinging and turning physically rather than relying solely on digital movement, to ease on potential for cyber-sickness.

3.2.3 Participants

After receiving approval from the Trinity College Dublin's School of Computer Science and Statistics Ethics Committee, participants were recruited through direct contact with the researcher, either via e-mail, or in-person at the place this study was conducted and written. In total, 14 participants were recruited, and all completed the study.

No compensation was offered to participants for their time, most were graduate students

at Trinity College Dublin, and the only age ranges represented were 18-24 and 25-34, specific ages were not recorded. 3 of the participants were female, and 11 were male. None of the participants had any experience with Volleyball prior to this study, and prior experience in VR ranged from never having used such devices before, to regular users.

3.2.4 Procedure

Upon arrival, each participant was greeted and asked to fill out the Informed Consent Form. After a short explanation of the procedure, participants were offered a bottle of water, and asked to put on the HMD and controllers, with help offered by the researcher for users with little prior experience with VR.

Participants then performed the task outlined above, with the researcher in the room to make private observations on exhibited behaviour, answer questions, and direct participants back towards the centre of the active space in case they moved closer to the surrounding walls. Special care was given not to reveal the full purpose of the study, beyond its title, to prevent participants from overthinking the influence of the crowds on their experience, as recommended by Murcia-Lopez et al. (2020). The researcher never actively engaged participants in conversation during gameplay, only intervening unprompted to direct participants towards a safe middle in the room.

During the gameplay phase, participants were permitted to take a break whenever they wanted or needed, which would pause the 20 minute minimum play time.

Following the experience, participants would cleanly exit the game, then proceed to answer a questionnaire relating to the experience, following which they were thanked for their participation. No study session lasted longer than 60 minutes overall.

3.2.5 Evaluation

The data collected towards this study is recorded in two separate prongs: Firstly, there is the crowd state history established during the participant task, which aims to showcase a trend in transitions. This history will also be used to establish a ranking of crowd states by final preference, due to the instruction to participants to close on their preferred configuration.

Questionnaire answers containing quantitative answers to certain questions on Likert scales, as well as open text answers to complete sections, will be leveraged to evaluate the hypotheses with quantitative answers, whilst also hopefully showcasing interesting

correlations, or in the case of open text answers, illustrate or nuance findings through direct user feedback.

Observations made by the researcher during participants' gameplay, such as particularly salient behaviours will also be used to illustrate or nuance findings, though they cannot be used to directly prove or disprove hypotheses, considering expressed behaviours don't always match inside states and may mirror other factors, e.g. the presence of other people in the space, or comfort in exchanging with the researcher or not.

3.3 Summary

In short, we have outlined the potential uses of a study outlining the effects of crowds on plausibility, to avoid breaks in presence during gameplay in VR sports application, which would lead to a loss in motivation and performance in athletes using such technology to support their training regime.

This potential aims to be addressed by this study by answering the question, whether trade-offs in crowd population meaningfully affects plausibility in a VR beach volleyball scenario, chosen for its relative absence of scrutiny in existing literature.

This question is addressed in a participant study featuring 14 young adults, within the age range of 18-34, without any prior experience in volleyball, playing a purpose-built game in which they would either serve, or react to balls played in from the other side of the net. During gameplay, they would adjust crowd density using a regularly appearing prompt, producing a history of crowd transition ending with a given participant's preferred state overall. Participants also answered a questionnaire to assess their reactions in a more quantitative fashion, with open text fields to relate any nuance or more expressive answers.

Chapter 4

Implementation

In the previous chapter, the task participants were asked to perform in order to measure plausibility of crowds was described. Performing this task, predictably, requires the existence of a platform to do so.

In a first section, the absence of a viable, commercially available product to do so is laid out, before the purpose-built game made towards this project is detailed in follow-up sections, broken down by individual components.

4.1 Absence of a commercially available alternative

For use of a standalone VR device, when looking for applications to download and use, the Meta app store is the main platform to use. A search was made to find games with the only keyword queried being 'volleyball', which produced only four results: "Volleyball Fever" Neuston AB (2021), "Volleggball" Pink Marmot Studios (2021), "BIG BALLERS '26" Doge Labs Inc. (2023), and "Clawball" ARVORE Immersive Games Inc. (2024).

The last two are games for a different sport altogether, tailored for basketball, and football respectively. Of the other two games, neither featured any visual crowds whatsoever, and the trailer of Volleyball Fever did not indicate any audio tracks to represent a disembodied crowd, while Volleggball includes these when points are scored.

This researcher neither had the skills nor the time to attempt to mod either of these games for the purposes of this project, therefore the decision was taken to make a volleyball game from scratch using a conventional game engine.

4.2 Overview of the Solution

The requirements for a game supporting the participants' task are for the game to take place in a suitable visual environment, enable classic interactions with the ball as accurately as possible to resemble their real counterparts, a baseline set of volleyball rules, and to feature a crowd that could be dynamically adjustable by way of a user interface (UI) field.

In order to accommodate the free play mode, there was also a need for a baseline tutorial phase, that also did not count points or propose the crowd adjustment panel. Finally, to increase engagement, a varied set of play situations was considered desirable.

For development, two game engines were considered for their status as industry-leading, as well as the researcher's prior experience in working with both. Unity Technologies (2026) was chosen over Unreal Engine Epic Games Inc. (2026) for the researcher's comfort in using the engine in prior projects, alongside Unity's comprehensive VR template packages that already implemented many VR features, such as grabbing and throwing, or the provision of a fully VR-rigged avatar with basic movement and interactions. The chosen version reflected the latest update at the time development started, Unity 6000.4.8f1.

Over the next few pages, the various components of the projects will be outlined in order of implementation, beginning with the template assets, followed by the crowds and game rules, then a quick primer on Unity's physics engine before going into ball physics and interactions, then closing off with any parts of components that didn't deserve a dedicated section.

4.2.1 Unity's VR[MP] template

Go into the features reused for the project:

- movement: Left-Right-Forward-Backwards via left joystick + teleport
- jump
- XRGrabInteractable and components
- XRInteractionSetup
- settings menu

might also dedicate a quick sentence to scene setup, either here or before starting the actual implementation to reference the assets used + procedurally generated terrain.

4.2.2 Stadium Crowd Generator

Virtual crowd from plugin AiKodex (2021), modified for:

a) randomised seating (! not at runtime, every participant had the exact same crowd), and

b) dynamic density adjustment. Audience was composed of a random choice between n different spectator prefabs, each assigned a distinct animation rooted in a different sentiment, and all wearing a different colour shirt.

Spectators were diverse* in both gender and ethnicity, and were recognisably Human but did not have any faces, thus couldn't have any facial expressions, any reaction was based solely on gestures.

+ a word on audio

4.2.3 Game Rules

- Ball landing point recognition (on or off court + side landed on)
- Movement boundaries as invisible walls
- ball boundaries
- Match flow (points system + crowd change prompt) and ball respawn
- ball lifetime states

4.2.4 Ball Physics

Quick tangent on Unity Physics, as much as I can find in documentation + Ball Material properties

4.2.5 Hitting Types

- Initial hope to leverage built-in physics engine crushed by non-consideration of controller velocity
- Attempts at velocity estimation based on positional difference: raw difference, smoothened difference, weighted smoothened difference + explain lack of reliability (frame length variability mostly) then transition to controller-based velocity reading

- Hit simulation by using hand colliders as triggers rather than physical objects → aim estimation + force 'balancing' based on trial and error
- hit classification

4.2.6 Other

- maybe a word on the plan to include bots? Scrapped for time concerns
- Incoming ball mechanic: Spawn at random point above net and aim for player at bounded random strength for variability in receiving height.

4.2.7 Performance

will get framerate and stuff like hardware/memory usage (?) today.

4.3 Summary

Every chapter aside from the first and last chapter should conclude with a summary.
structure of chapter:

- Overview (Detail how there wasn't a satisfactory, commercially available VR VB app for the study, so had to make my own. List requirements.)
- Technical components
 - VRMP template + assets
 - Crowds - visuals, audio, crowd levels
 - match management + implemented rules + ball lifetime
 - short section on ball physics (citing Unity documentation on basic physics)
 - Hitting physics - iterations from built-in physics, velocity approximation, then sensor collection
 - hitting mechanics - aim estimation, hit strength modifiers, ways of application.
 - Incoming ball mechanic.
- Summary of the section

Chapter 5

Evaluation

The Evaluation chapter should present a comparison of the work that forms the basis of the dissertation and existing work. At a higher level, it should demonstrate an awareness of the relationship of the dissertation work to the research area that it is based in.

5.1 Experiments

In the case where experiments have been carried out, the experimental setup and the values that were defined for the variables need to be presented in a table e.g. table 5.1.

Column 1	Column 2	Column 3
Row 1	Item 1	Item 2
Row 2	Item 1	Item 2
Row 3	Item 1	Item 2
Row 4	Item 1	Item 2

Table 5.1: Caption that explains the table to the reader

5.2 Results

Figures that present results such as figure 5.1 need to display descriptions of the axes, the units and scales of the measurements, statistical values, etc. Where measurements were taken from experiments, error bars or confidence intervals need to be provided to give the reader an indication of the spread of the measurements.

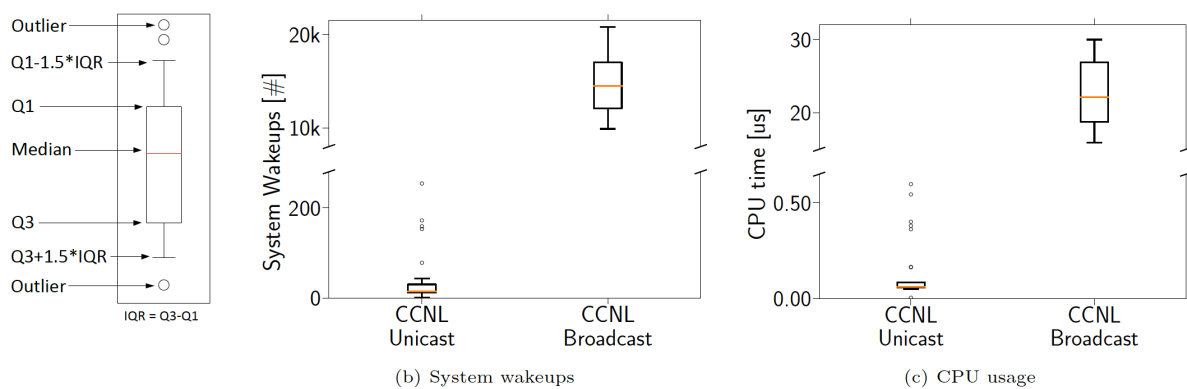


Figure 5.1: Long caption that describes the figure to the reader

5.3 Summary

Every chapter aside from the first and last chapter should conclude with a summary that presents the outcome of the chapter in a short, accessible form.

Chapter 6

Conclusions & Future Work

This chapter should summarize the work presented in the dissertation and discuss the conclusions that can be drawn from the work and the results presented in chapter 5.

6.1 Limitations

- VB in VR exposes MANY limitations of iVR in general:
 - complete absence of haptic feedback
 - dangerousity from lack of vision of feet/ground
 - need for micro-detail recognition (ex. for blockers, any indication, e.g. eye or head movement, hand orientation and signals, from the setter where the ball will go counts)
 - very complex ball handling compared to other sports (uniquely so)
 - complexity of implementation to a realistic level (many nitpicky rules e.g. double touching, holding the ball for too long during a set, setting foot on the other side of the net by accident etc.)
- simplicity of implementation + faulty aim estimation & hit classification
- questionnaire not nearly good enough: Open text fields not leveraged at all, crowd influence should have been measured from negative influence to positive with neutral/unaffected as a mid value instead of extreme
- usefulness of transition matrix? Yet to look at it more closely but on face value it seems like a bit of a pointless metric idk

6.2 Future Work

The section may present a list of items that were beyond the scope of the dissertation.

Game improvements:

- use of controller-free controls (e.g. hand-tracking, maybe full-body or foot-tracking for more holistic games) - e.g. for specialised systems through a real-time MoCap setup → could be paired with haptic gloves to simulate ball physics on hands!
- Better hit classification (esp. from one hand), refining aim recognition for specific mechanics (e.g. dig particularly)
- including actual opponents rather than just reacting to balls flying in

Study direct improvements:

- use as a learning experience, e.g. using this to teach complex IRL mechanics (e.g. setting) and measuring how well that translates to improvement on the pitch.
- more in-depth probing for results (e.g. sentiment analysis on interviews or similar, more open responses, as lauded by Slater in Slater et al. (2023) to pick up on a much larger depth of information than a 10 point scale ever could)

Tangential directions for further research:

- More alive/reactive crowds (with more accurate levels of engagement, maybe recorded from events of the actual sport; also more individual shouts, more reactivity to on-pitch happenings (oohs and ahhs with bigger events e.g. blocks), or wider range of emotion - cheering, booing, jeering, chants, boredom/relative quiet etc.), e.g. using researched crowd behaviour models like the one used for Glémarec et al. (2021).
- more detailed investigation into human factors (what kind of trait reacts the most to the taunting spectator, what kind of trait is associated with complete indifference from the crowd etc.)
- add more factors to crowd for participant control, e.g. detach audio from crowd level (what if lv 2 audio was more accurate with a fuller crowd for example).

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Appendix

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